

# CSS Styling & Layout

ResNet Academy | Frontend Development

## Specificity and the cascade

| Selector                             | Score (id, class, type) |
|--------------------------------------|-------------------------|
| <code>p::before</code>               | (0,0,1)                 |
| <code>.card[type=email]:hover</code> | (0,1,0)                 |
| <code>.card p</code>                 | (0,1,1)                 |
| <code>#nav a</code>                  | (1,0,1)                 |
| Inline <code>style=""</code>         | beats all selectors     |
| <code>!important</code>              | beats inline. Avoid     |

Winner: higher specificity, otherwise the later rule. Text properties inherit. Box properties do not.

## Box model

```
*, ::before, ::after { box-sizing: border-box; }
```

margin (outside) > border > padding > content  
margin: top right bottom left (clockwise)  
margin: 0 auto (centre a block with a width)  
Vertical margins of adjacent blocks collapse to the larger one

## Units

| Unit               | Relative to       | Use                                   |
|--------------------|-------------------|---------------------------------------|
| <code>rem</code>   | Root font size    | Type and spacing                      |
| <code>em</code>    | Element font size | Component-relative spacing            |
| <code>%</code>     | Parent size       | Fluid widths                          |
| <code>vw vh</code> | Viewport          | Full-screen sections                  |
| <code>ch</code>    | Width of 0        | <code>max-width: 65ch</code> for text |
| <code>fr</code>    | Grid free space   | Grid tracks                           |

## Positioning

| Value                 | Behaviour                                               |
|-----------------------|---------------------------------------------------------|
| <code>static</code>   | Normal flow (default)                                   |
| <code>relative</code> | Normal flow, offset allowed, anchors absolute children  |
| <code>absolute</code> | Out of flow, placed against nearest positioned ancestor |
| <code>fixed</code>    | Placed against the viewport, stays while scrolling      |
| <code>sticky</code>   | Relative until a scroll threshold, then sticks          |

## Flexbox

```
.container {
  display: flex;
  flex-direction: row | column;
  justify-content: flex-start | center | space-between | space-around; /* main axis */
  align-items: stretch | center | flex-start; /* cross axis */
  flex-wrap: nowrap | wrap;
  gap: 1rem;
}
.item { flex: 1; } /* flex: grow shrink basis */
.item { align-self: center; } /* one item only */
```

## CSS Grid

```
.grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr); /* or 1fr 2fr, or 240px 1fr */
}
```

```

    gap: 1rem;
}
.wide { grid-column: 1 / 3; } /* span from line 1 to line 3 */
.layout { grid-template-areas: "header header" "side main"; }
.layout > header { grid-area: header; }

```

**Key idea.** Flexbox is one-dimensional and content-driven. Grid is two-dimensional and layout-driven. Combine them: Grid for the page, Flexbox inside components.

## Useful selectors

| Selector                             | Meaning                                                 |
|--------------------------------------|---------------------------------------------------------|
| <code>.a .b</code>                   | <code>.b</code> anywhere inside <code>.a</code>         |
| <code>.a &gt; .b</code>              | <code>.b</code> directly inside <code>.a</code>         |
| <code>h2 + p</code>                  | The <code>p</code> immediately after an <code>h2</code> |
| <code>a:hover a:focus-visible</code> | Pointer over / keyboard focus ring                      |
| <code>li:nth-child(even)</code>      | Every second <code>li</code>                            |
| <code>input[type=email]</code>       | Attribute match                                         |